

Problem I – Inner Product

You are given a sequence of positive integers $a_1, a_2, \dots, a_n$ and a string s of length $n - 1$. For each i from 1 to $n - 1$:

- if s_i is equal to `<`, then $b_i < b_{i+1}$;
- if s_i is equal to `=`, then $b_i = b_{i+1}$;
- if s_i is equal to `>`, then $b_i > b_{i+1}$.

Your task is to choose a sequence of positive integers $b_1, b_2, \dots, b_n$ satisfying all these relations. Among all valid sequences, minimize

$$a_1b_1 + a_2b_2 + \dots + a_nb_n.$$

For the given constraints, it can be proved that the optimal sequence is unique.

Input

The first line contains one integer n ($2 \leq n \leq 200'000$).
The second line contains n integers $a_1, a_2, \dots, a_n$ ($1 \leq a_i \leq 1'000'000$).
The third line contains a string s of length $n - 1$, consisting only of the characters `<`, `=`, and `>`.

Output

Print the minimum possible value of

$$a_1b_1 + a_2b_2 + \dots + a_nb_n$$

in the first line.
In the second line, print the unique optimal sequence $b_1, b_2, \dots, b_n$.
It can be proved that, for the given constraints, the minimum value always fits in a signed 64-bit integer.

Sample input 1 6 3 1 4 1 5 9 <=>><	Sample output 1 43 1 3 3 2 1 2
Sample input 2 2 2 7 <	Sample output 2 16 1 2
Sample input 3 8 8 2 2 8 3 7 4 6 =<<=>>=	Sample output 3 71 1 1 2 3 3 2 1 1

Notes

The first sample requires

$$b_1 < b_2 = b_3 > b_4 > b_5 < b_6.$$

The sequence 1 3 3 2 1 2 satisfies all relations, and its cost is

$$3 \cdot 1 + 1 \cdot 3 + 4 \cdot 3 + 1 \cdot 2 + 5 \cdot 1 + 9 \cdot 2 = 43.$$